Giant Ionic Structures

16/10/2024

Keywords: ionic bonding, electrostatic, ions, conductor, melting point, boiling point

DO NOW: Let's do a quiz about your previous lesson. Quiz 1 □ 5

Revision

16/10/2024

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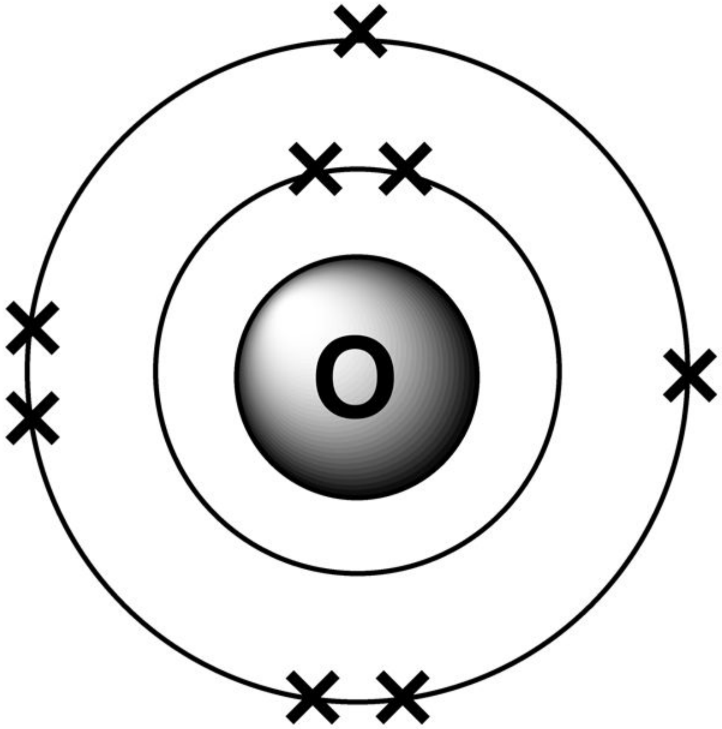
DO NOW: a) Draw this oxygen as an ion

b) What is the electron configuration of the atom and ion



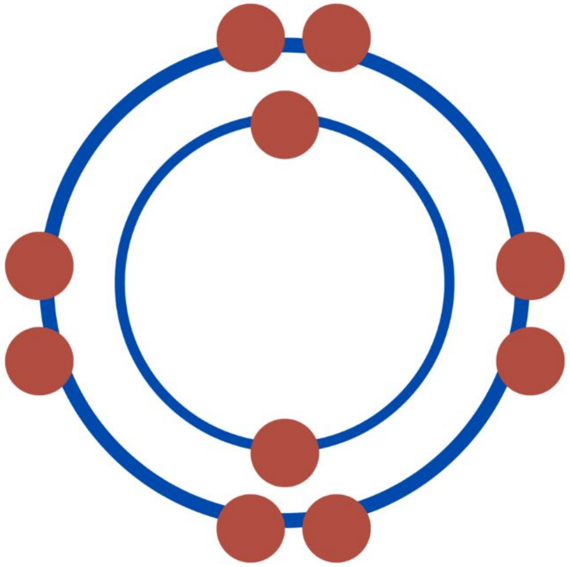
LOb: Understand the arrangement and properties of giant ionic structures.

Keywords: ionic bonding, electrostatic, ions

Learning Outcomes:

- Describe a giant lattice structure
- Explain properties of giant ionic structures
- Write formulas for ionic bonds

OCTET RULE



Did you know… the octet rule is that atoms lose or gain electrons to have full outer shells of 8 electrons.

Here are some giant ionic compounds

cooper sulphate ($CuSO_4$)

Metal ion

Non-metal ion

Ionic bonds form a

<u>giant lattice structure</u>

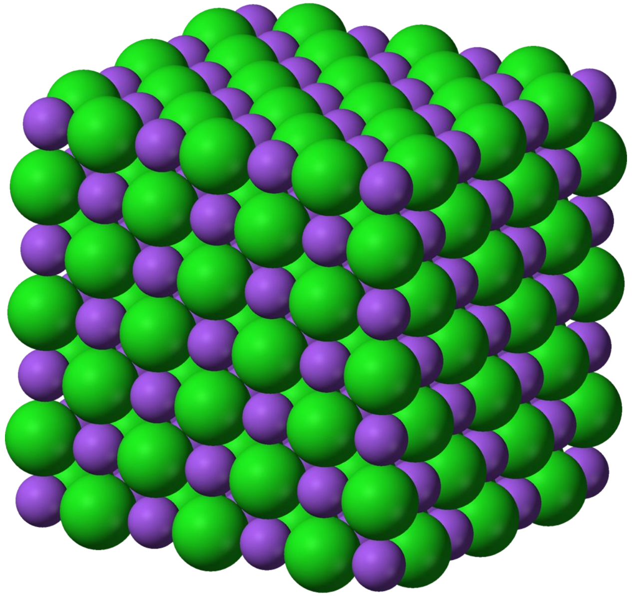
sodium chloride ($NaCl$)

Metal ion

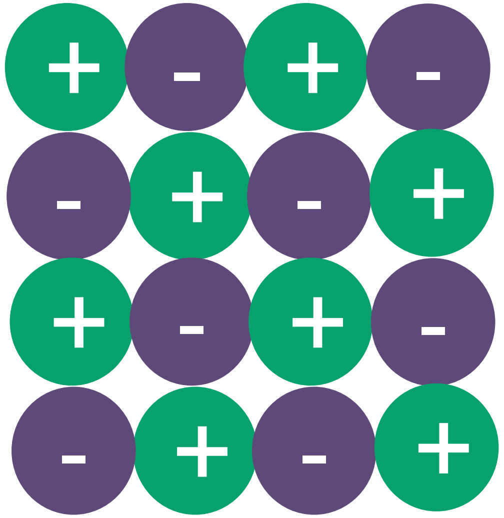
Non-metal ion

Giant lattice structure

A repeating pattern of positive and negative ions.

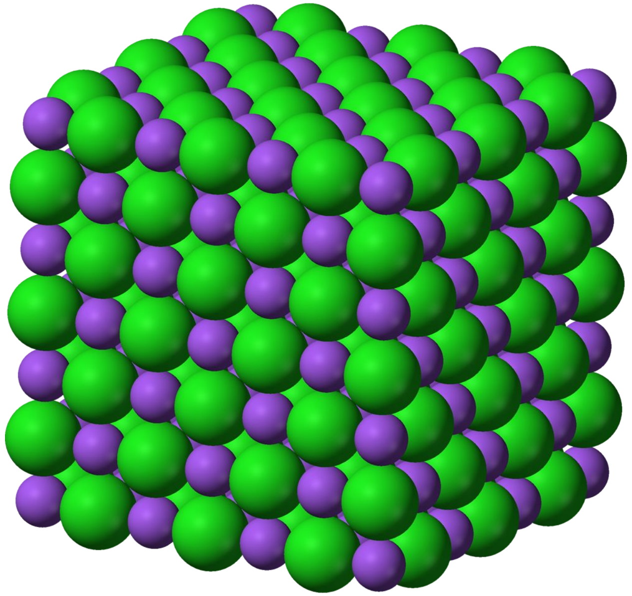


3D view



2D view

Ionic compounds have **strong electrostatic forces** of **attraction** between ions **acting** in **all directions.**



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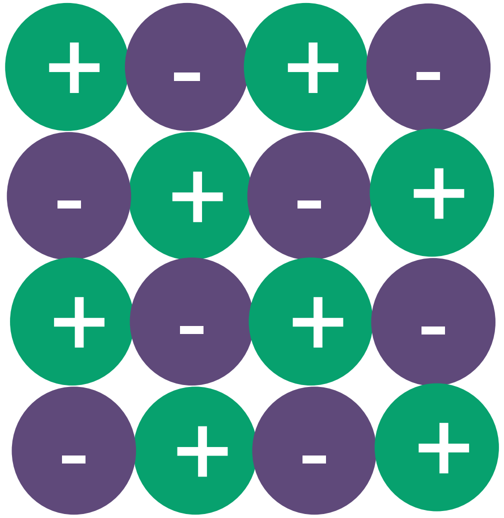
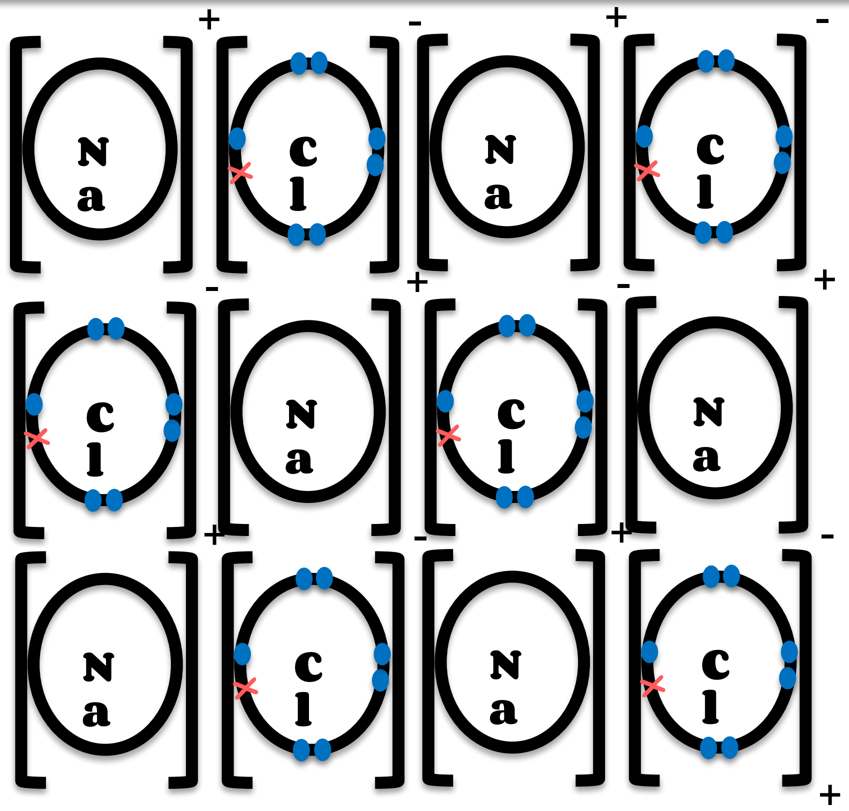
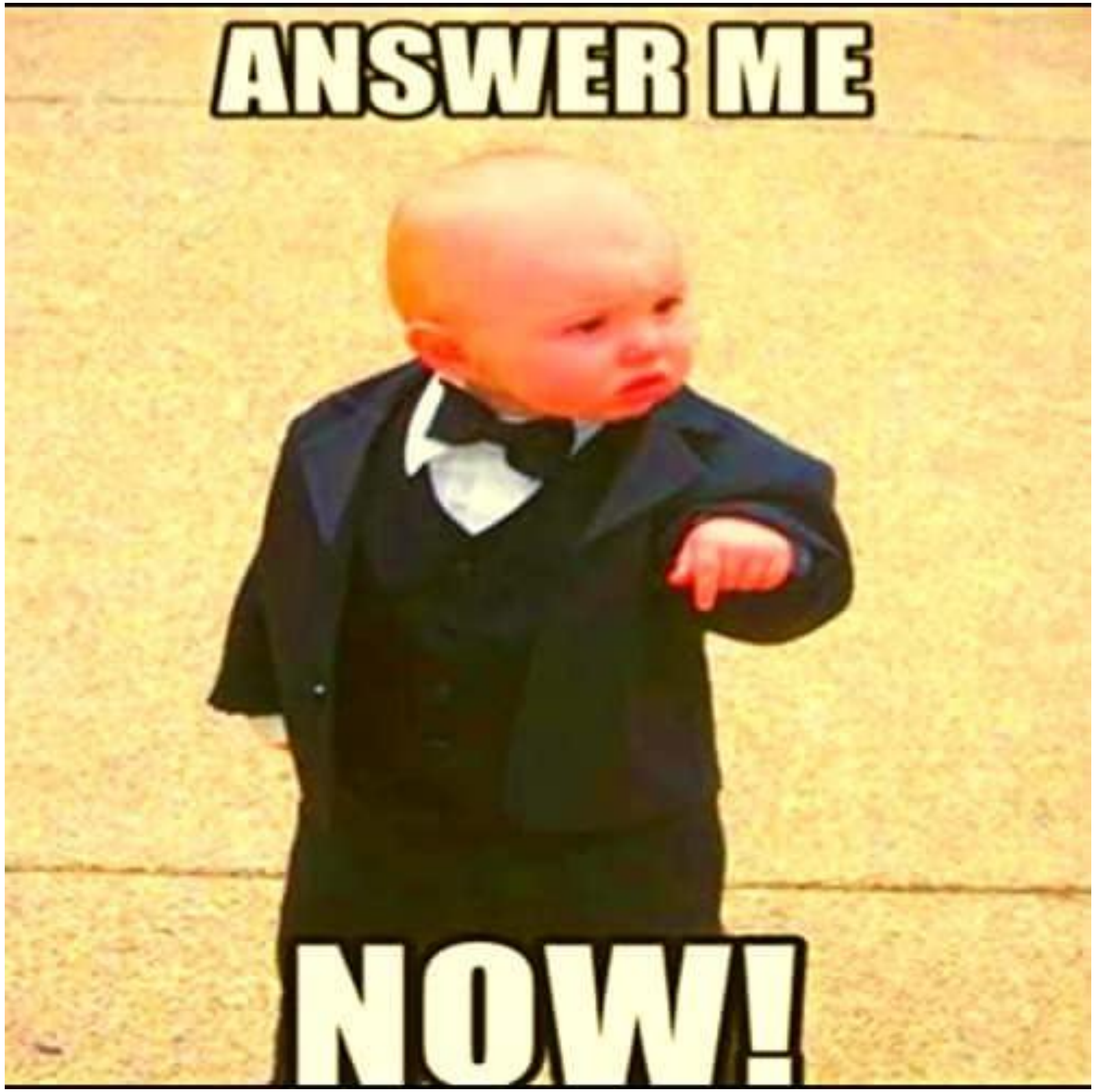


2D view



2D showing ions

YOU DO: Complete the questions in you work book



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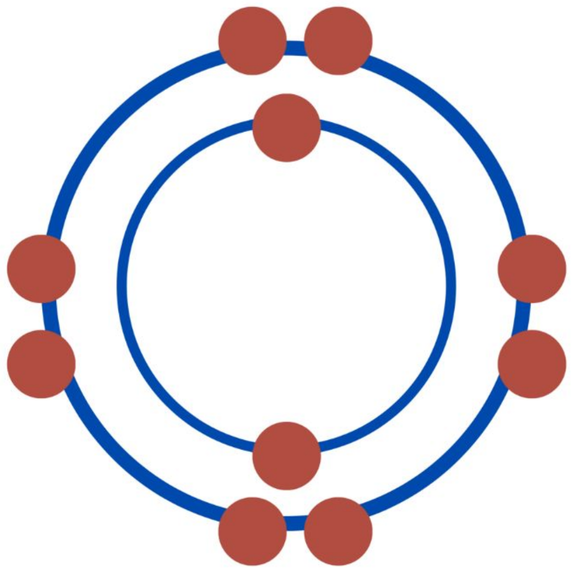

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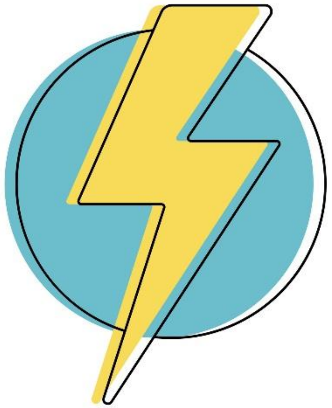
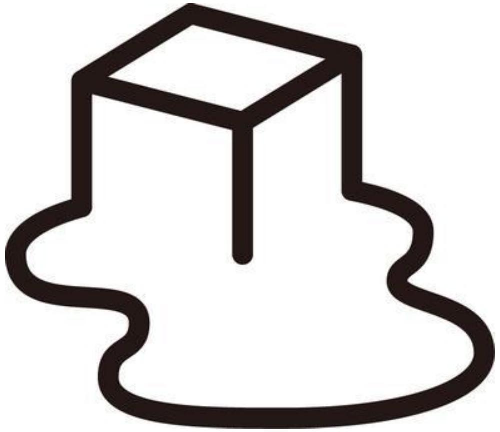
Giant Ionic Bond Properties



Electrical Conductor



High m.p & b.p

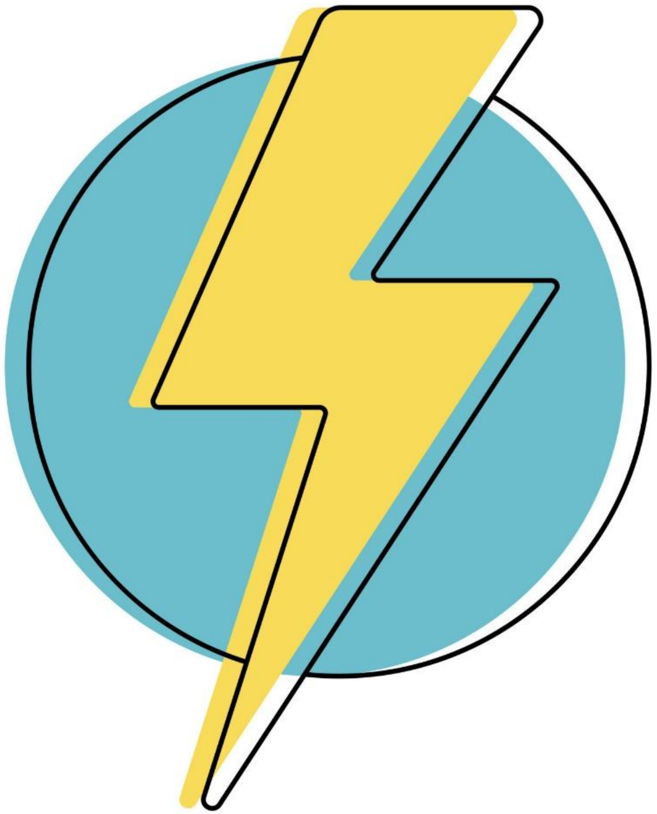
YOU DO: Watch the video of ionic bond properties

WATCH AND

LEARN

AFTERALL KNOWLEDGE IS

POWER!

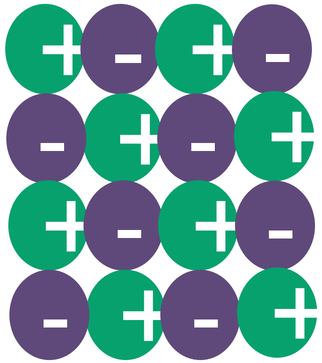


Review: Conductor

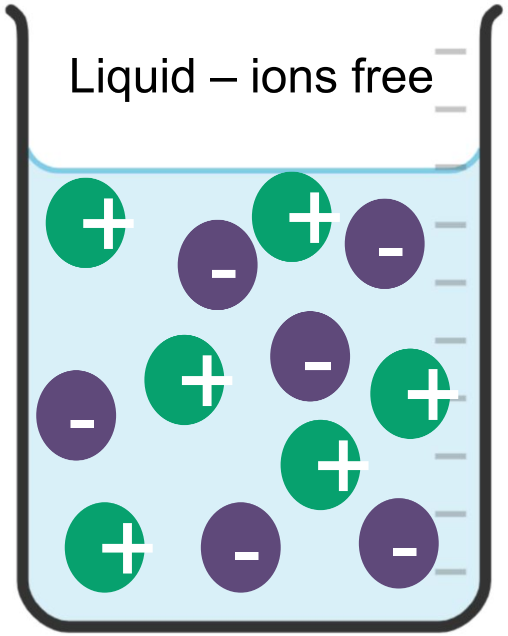
FACT: Ionic bonds have ions that are **free to move** when **dissolved** or **molten**.

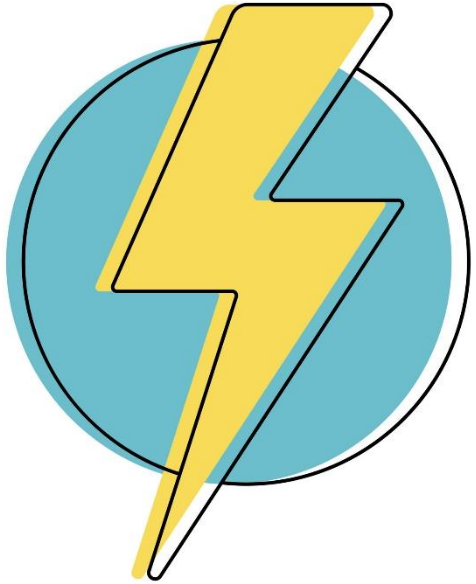
So…

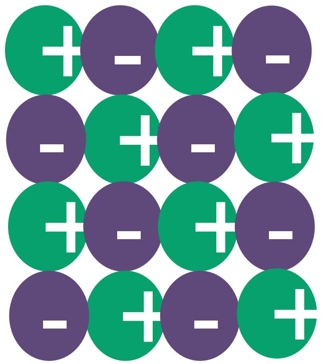
STATEMENT: the **ions** can **carry the charge** through the liquid.



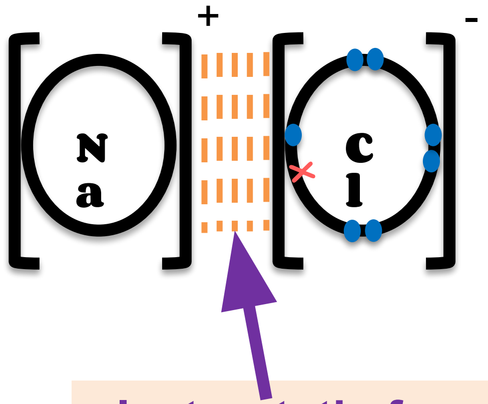
Solid – ions not free



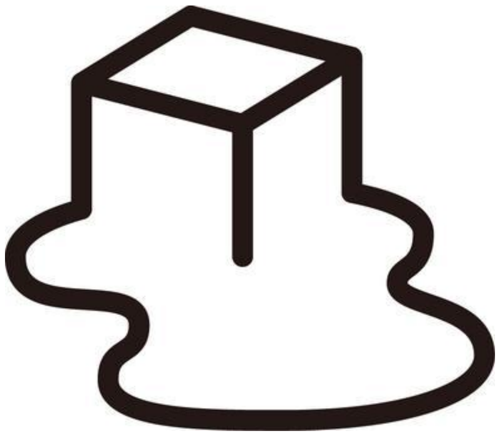




To melt or boil this structure means to loosen the ions in this structure.



electrostatic force holding the ions together is very strong.



High m.p & b.p

thermal energy

Key words

strong

electrostatic force

energy

break

Review: Melting & boiling point

FACT: Ionic bonds have **strong electrostatic forces of attraction** between the ions,

So…

STATEMENT: a **large** amount of **thermal energy** is required to **break** the force.

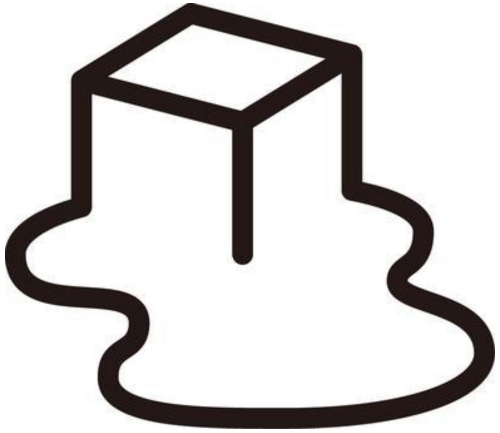
Key words

strong

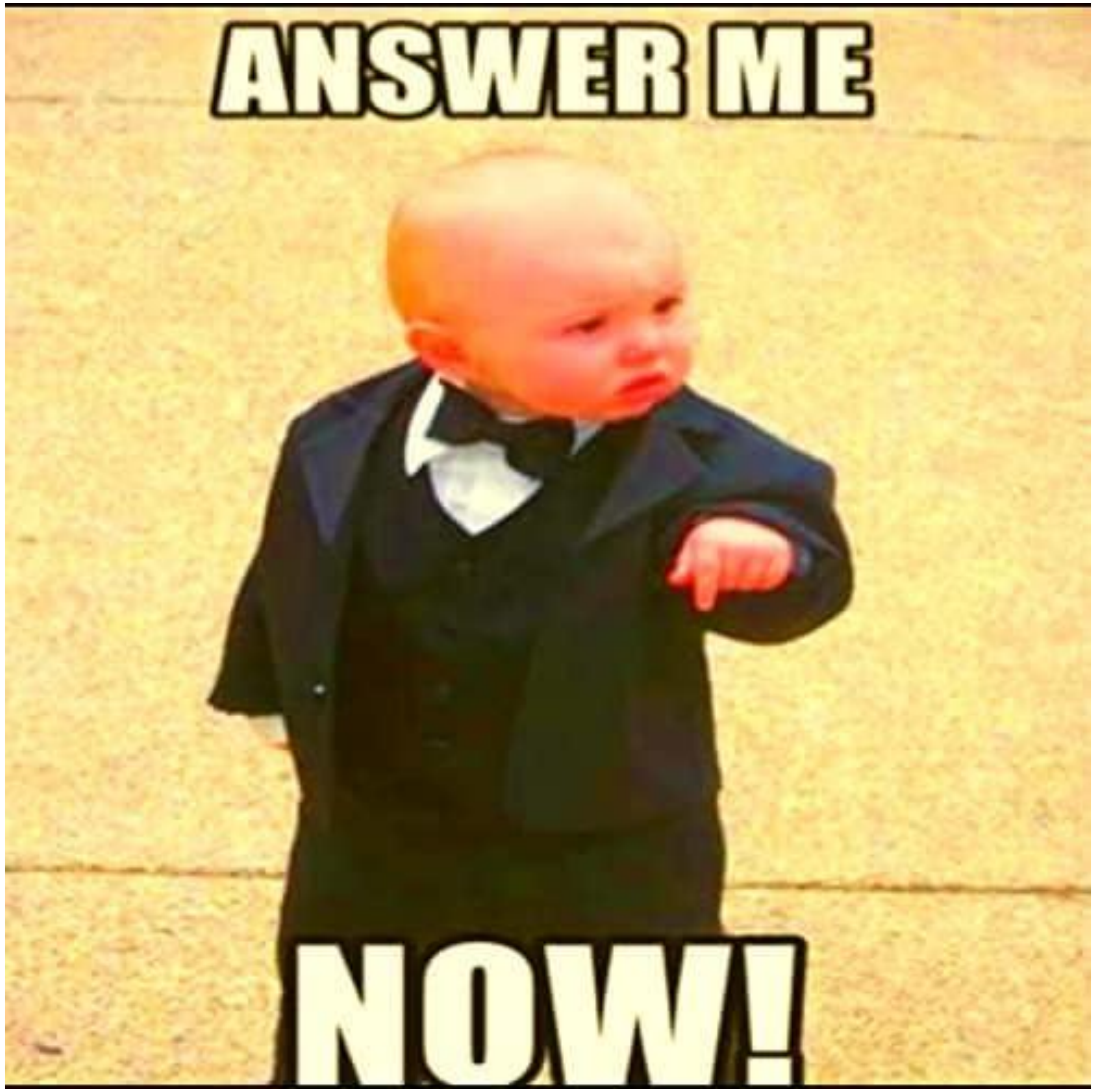
electrostatic force

energy

break



YOU DO: Complete the questions on your sheet.



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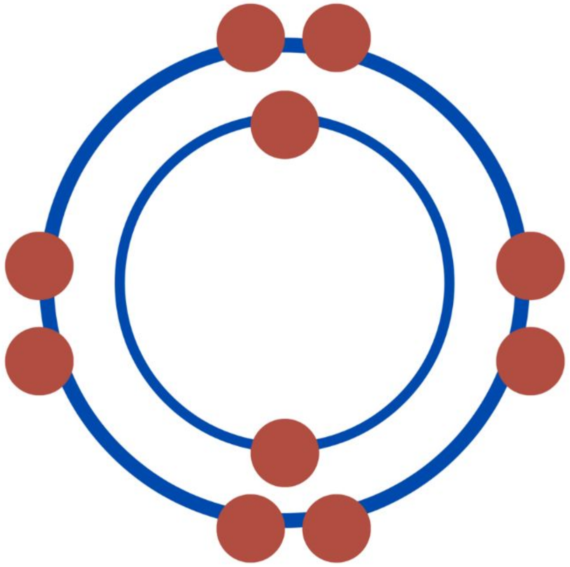

Did you know… the octet rule is that atoms lose or gain electrons to have full outer shells of 8 electrons.

When sodium reacts with oxygen to create an ionic bond what is its formula?

STEP 1 Find the formula for each element on the periodic table

11 22,990	8 15,999
Na	**O**
Sodium	Oxygen

STEP 2 Find the charge of each element
Metals^{+} group number tells you the size
Non-metals^{-} 8 – group number tell you the size

Na^{1+} O^{2-}

STEP 3 SWAP CHARGES

Na^{2-} O^{1+}

STEP 4 Place as subscripts to create formula Na_2O

When aluminium reacts with Fluorine to create an ionic bond what is its formula?

STEP 1 Find the formula for each element on the periodic table

13 26,982	9 18,998
Al	F
Aluminium	Fluorine

STEP 2 Find the charge of each element
Metals^{+} group number tells you the size
Non-metals^{-} 8 – group number tell you the size

Al^{3+} F^{1-}

STEP 3 SWAP CHARGES

Al^{1-} F^{3+}

STEP 4 Place as subscripts to create formula AlF_3

YOU DO: Write the formulas for these ionic compounds

a) strontium (Sr^{2+}) nitride (N^{3-})
b) magnesium oxide
c) lithium oxide
d) calcium bromide
e) aluminium nitride
f) potassium nitride

EXT:

- ❑ lithium sulphate
- ❑ calcium carbonate
- ❑ aluminium nitrate

Nitrate = NO_3 = 1- charge

Sulphate = SO_4 = 2- charge

Carbonate = CO_3 = 2- charge

YOU DO: Complete the sheet questions

Review

a) strontium (Sr^{2+}) nitride (N^{3-}) Sr_3N_2
b) magnesium oxide MgO
c) lithium oxide Li_2O
d) calcium bromide $CaBr_2$
e) aluminium nitride AlN
f) potassium nitride K_3N

EXT:

- ❑ lithium sulphate Li_2SO_4
- ❑ calcium carbonate $CaCO_3$
- ❑ aluminium nitrate $Al(NO_3)_3$

Nitrate = NO_3 = 1- charge

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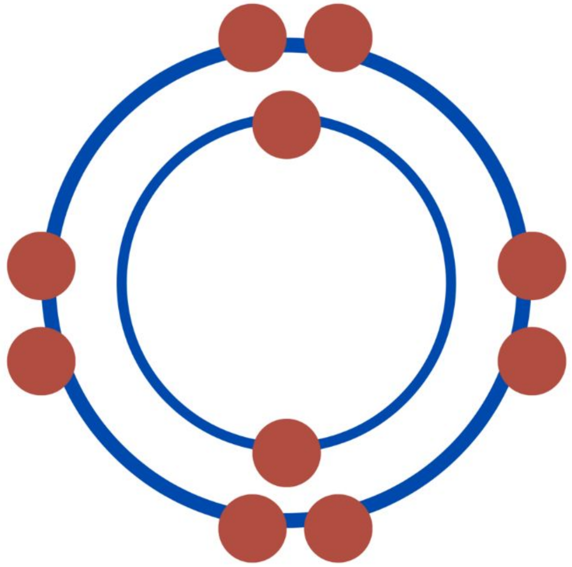

Learning Outcomes:

- ✓ Describe a giant lattice structure
- ✓ Explain properties of giant ionic structures
- ✓ Write formulas for ionic bonds

Did you know… the octet rule is that atoms lose or gain electrons to have full outer shells of 8 electrons.

keyfacts

- Ionic bonds form a **giant lattice structure.**
- Between ions in the structure there are **strong electrostatic forces**
- Properties: 1) **High melting** and **boiling point** 2) **Conductor** only when **molten** or **dissolved**
- To write the formula **find the charges** of each element, then **swap charges**, then **write as subscripts.**

Plenary

Which of these substances in the table are ionic? Justify you answer.

Substance	Melting point (°C)	Boiling point (°C)	Does it conduct electricity?		
			Solid	Liquid	Solution (aq)
A	650	1107	yes	conducts	insoluble
B	114	184	no	no	no
C	801	1467	no	yes	yes
D	2040	2980	no	yes	insoluble
E	119	445	no	no	insoluble
F	1610	2230	no	no	insoluble

Plenary

C = High m.p & b.p and conducts only when in solution or molten.

D = High m.p & b.p and conducts only when molten.

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Ionic Bonding Practical

16/10/2024

Keywords: ionic bonding, atoms, compounds, mixtures, electrons, negative, positive

DO NOW: a) Write the substances that are ionic bonds

b) Write the symbol and charge of ions found in **Na_2O**

Sodium bromide

Ammonium (NH_3)

Hydrochloric acid

water

Aluminium lithium

Carbon dioxide

Calcium nitride

LOb: Understand properties of ionic bonds

Keywords: conductor, sodium chloride

Learning Outcomes:

- Describe properties of ionic bonds
- Investigate the conductivity of ionic bonds

Did you know… This quote is wrong. Scientifically the strongest bonds are called ionic and covalent bonds. They happen when certain elements chemically join together

Required Practical

To investigate the conductivity of solid and dissolved ionic compounds.

I predict...

Equipment list

Pot of NaCl
3 wires
Power pack (6V)
Lamp
Electrode holder

Expectations

- Follow only the instructions
- Stay in your group
- Hands to yourself
- Record the results as they happen

Job roles

A: Equipment monitor in

C: Equipment set up instruction step 1

D: Water filler and salt mixer

B: Equipment monitor out

E: Step 1a

F: Controls power pack

Method

1b) Connect the powerpack to the lamp with 1 wire.
Connect the lamp to the electrode holder with the 2nd wire.
Connect the electrode holder to the powerpack with the 3rd wire

2. Place the electrodes into the pot of salt
3. Turn on the power pack and observe.
4. Write what happened into the table. Turn off power pack
5. Add 150 cm^3 of water to the beaker and pour half the pot of NaCl into the beaker
6. Place the electrodes into the beaker
7. Turn on power pack. Observe a record results

STEP: 1a. Connect only the lamp to the power pack and check the light bulb is working

Results

State of matter	Observation of light bulb and solution
Solid	
Solution (dissolved)	

Review

The practical was successful | unsuccessful.

We saw…

This agreed | disagreed with my prediction

The reason it might have been unsuccessful was…

Q1. Write the ions found in the solution you made

Q2. Explain why ionic bonds conduct only when molten or dissolved

Q2. Draw the reaction of sodium reacting with chlorine